Client Report

(CCB – Risk Program Senior Associate)

Income Prediction & Customer Segmentation Using CPS Data

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1. Executive Summary:

The goal of this study was to predict individual income levels and identify high-value customer groups using the U.S. Census microdata for marketing purpose and product targeting by JPMorgan Chase's Consumer and Community banking division. In this report we have leveraged feature engineering, machine learning, statistical analysis and clustering based segmentation techniques to identify these potential customer groups.

Our exploratory data analysis (EDA) revealed a highly imbalanced income label (very common in fraud cases as well). This finding helped us decide several modeling steps, such as, class weighting, prioritizing the use of holistic metrics (ROC, AUC and Precision-Recall) instead of accuracy. Our findings suggest that **education**, **mean weeks worked in a year**, **occupation**, **marital status**, and **wage per hour** displayed **strong correlation** with income.

In the second phase, a clustering framework was developed using **k-means algorithm** to segment the adult population into separate groups for marketing purpose. We identified four distinct customer segments. One cluster, characterized by median age 39, higher education, full-year employment, professional occupations, married households emerged as a high value segment. Our feature importance analysis also confirmed that *work intensity* and *wealth* related attributes are the strongest determinants of high-income status.

Therefore, this study provides actionable business insights into income prediction, customer heterogeneity and financial product targeting, enabling a more efficient and precise client acquisition.

2. Data Understanding:

2.1 Dataset overview:

Our dataset [1] contains ~199,000 individual-level records and over 40 socioeconomic variables. The target variable indicates whether an individual earns more than $50k annually, with a strong class imbalance. Initial descriptive statistics revealed that many categorical fields contain "*Not in universe*" values, wealth variables such as capital gains and dividends are highly right-skewed and work-related variables exhibit wide variation in labor force participation. These insights informed our data cleaning, feature engineering and modeling strategies used throughout the project.

2.2 Exploratory Data Analysis:

In this section, we identify drivers of income, tackle class imbalance and prepare a downstream preprocessing pipeline for modeling and segmentation. Additionally, the descriptive EDA statistics were computed with respect to survey weights to ensure accurate population representative insights.

2.2.1 Income Distribution & Class Imbalance:

Weighted estimates revealed that only a small minority of individuals in the population earn more than $50,000 annually. This weighted probability of high income was significantly lower than the raw dataset proportion, which confirmed a severe class imbalance. To tackle this, we used synthetic minority oversampling technique (SMOTE), weight scaling in XGBoost and evaluated model performance using ROC-AUC and Precision/Recall over accuracy. **Fig 1.** depicts this class imbalance.

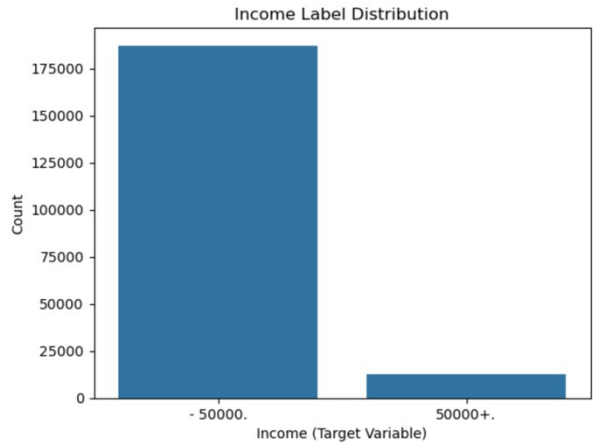

Figure 1. Bar plot to visualize label class imbalance

2.2.2 Education as a Primary Driver of Income:

In Fig. 2, we can observe a correlation between educational attainment and income. Individuals with advanced degrees showed weighted probabilities exceeding **50%–55%**, while those with only high school or lower education had probabilities closer to **0–10%**. Hence, education is one of the strongest socioeconomic predictors of income.

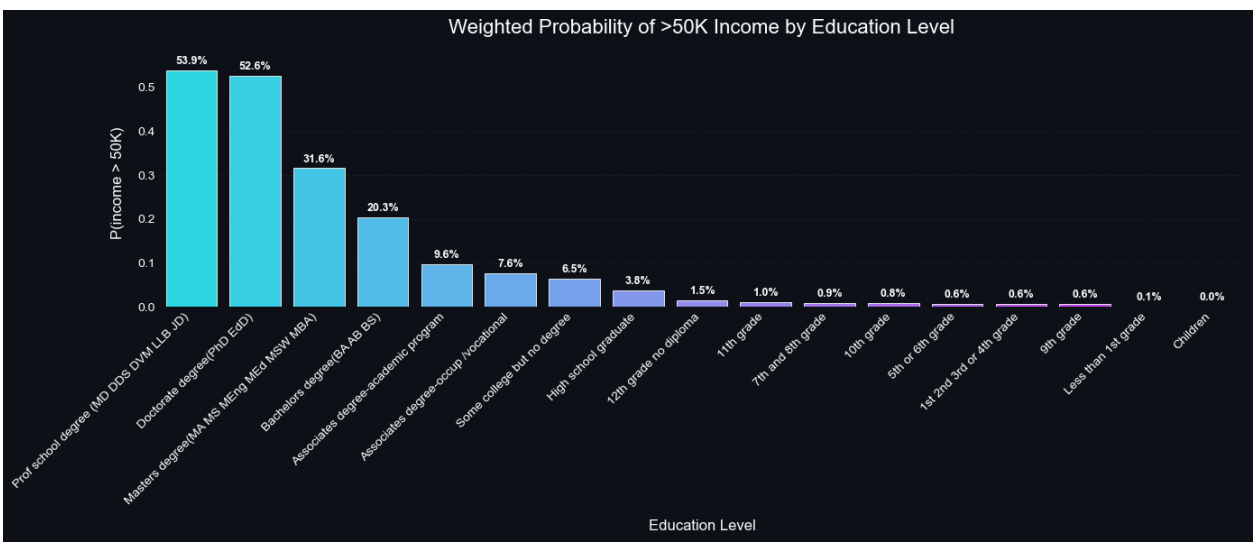

Figure 2. Weighted Probability of >$50K income by Education Level

Our main learning from this chart was to conduct an **Ordinal Encoding** of education column during feature modeling and inclusion of education level as a core clustering feature. This could prove a strong anchor for customer segmentation.

2.2.3 Occupational Segmentation of Income (Fig 3):

This was an interesting insight because the occupational patterns revealed clear stratification in earning potential. Findings are listed below.

a) Executive and Professional specialty roles showed the highest probabilities of earning >$50K.
b) Sales and Service roles formed a middle tier with moderate probabilities (~10%–20%).

c) Low skill occupations, such as handlers, machine operators, farming etc., registered very low probabilities and often below 3%.

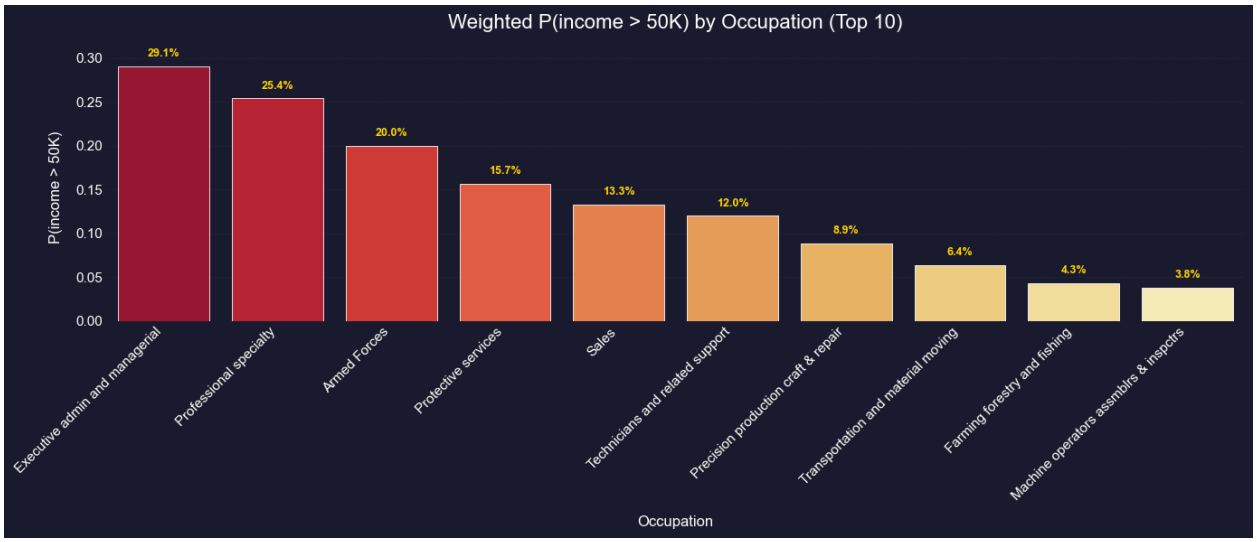

Figure 3. Weighted probability of >$50K income by occupation level

These findings influenced my modeling strategy in two ways. First, it encouraged me retaining occupational codes despite their high cardinality and second, prioritizing One-Hot Encoding instead of collapsing categories, preserving important economic distinctions.

2.2.3 Age as a Nonlinear Predictor of Income:

Age and income patterns demonstrated a nonlinear lifecycle effect, as it is seen with the trendline in **Fig. 4**. Young individuals (under 25) had near-zero probability of high income, whereas income probability climbs steadily through ages **35 to 55**, peaking at ~**17%** (also validated below in our segmentation results). However, it declines sharply after age 60, consistent with retirement trends. This plot validated the use of age, as a continuous predictor and justified not binning age for modeling.

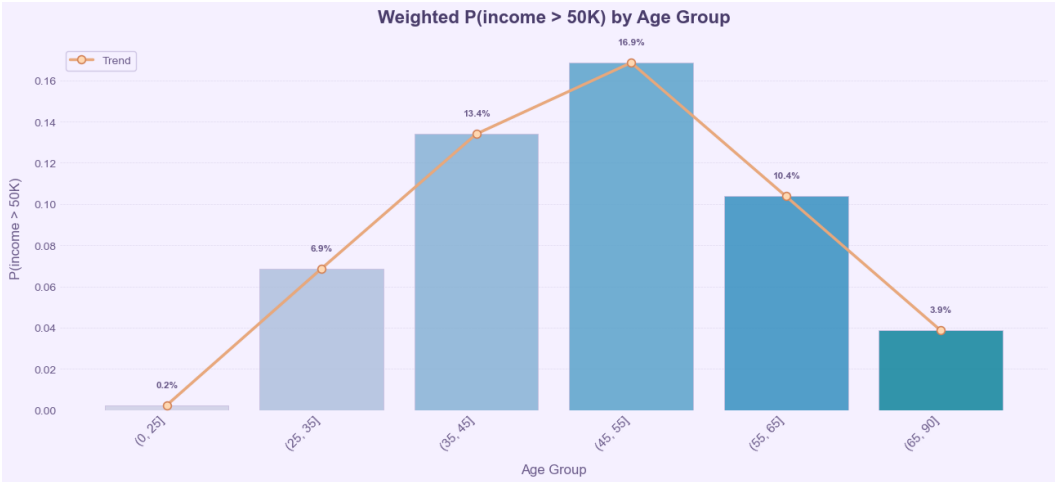

Figure 4. Trend analysis of Age with respect to Income

Summary of Statistical Findings:

In the above plots, we observed that across features like education, occupation and age, a consistent theme emerged: high income is driven by human capital (education), wealth indicators (capital gains), specific age bands, and the career type (occupation of the population group).

3. Data Cleaning and Preprocessing:

3.1 Handling Missingness and Structural Categories:

Categorical 'NaN' values were replaced with 'unknown' and numerical 'NaN' were replaced with median of the column, because it is robust to the skew of outliers. This dataset also includes responses such as *"Not in universe"*, reflecting individuals not eligible for certain labor or migration questions. Rather than treating these as traditional missing values, these responses were preserved as meaningful categorical indicators, especially for:

A) Employment-related attributes (for example, reason for unemployment, labor union status)

B) Family/household fields (for example, household summary, migration status)

3.2 Feature Standardization:

Numeric features exhibited substantial variation in scale. Capital gains peaked at 99,999, but wage per hour reached 9,999. To tackle this, we applied gaussian scalers achieving mean 0 and standard deviation 1 on the dataset. This step is also crucial for segmentation because K-Means clustering relies on Euclidean distance.

3.3 Data Encoding and Feature Engineering:

Ordinal Encoding (Education, Marital Status):

The column education was mapped to an ordered 1–16 scale, indicating increasing academic attainment. This also improved clustering and modeling accuracy by allowing the algorithm to learn income gradients associated with educational hierarchy (correlation shown in EDA section). However, marital status was categorized into simplified ordinal values representing household stability:

- 2 = Married
- 1 = Divorced/Separated
- 0 = Never married

These mappings reflect socioeconomic patterns relevant to financial behavior and income stability.

One-Hot Encoding (Occupation, Industry, Citizenship, etc.)

Categorical variables were one-hot encoded to avoid introducing false ordinal relationships and to preserve fine-grained occupational distinctions.

3.4 Tackling Class Imbalance:

Synthetic Minority Oversampling Technique (SMOTE) [3] artificially generates new minority-class samples by interpolation. SMOTE features were used for Logistic regression (Baseline) and Random Forest (Model 1). For the XGBoost algorithm positive class weighting was used.

Class Weighting: This technique maintains true data geometry while penalizing misclassification of the minority class. Its logical intuition is displayed in Eq. 1:

$$pos_weight = \frac{negative\ samples}{positive\ samples} \quad (1)$$

In our case, positive weight came up to be **15.11** units.

4. Predictive Modeling:

This section explains the modeling strategy used to identify groups of people earns more or less than $50K. The goal is to develop an interpretable and explainable model that could support business decision-making, along with enough evidence to justify the results, especially in scenarios involving customer segmentation. We started with a simple baseline and later moved to other complex architectures.

4.1 Model Architectures:

a) **Logistic Regression (Baseline):** Logistic regression provides a transparent and linear decision boundary with coefficients that map directly to feature importance though a Logit function. We chose this because it is faster to implement, easy to audit and therefore aligns with financial regulatory expectations around explainability.

   Architecture configuration: We leveraged L2 regularization along with maximum iterations of 4000 for convergence. Also, since it was a baselined, it was trained on SMOTE balanced and standardized numeric features

b) **Random Forest (Model 1):** Random Forests can model complex interactions and learn better correlations, for example, how age aligns with occupation due to a better education. It is also robust to noise and provide feature importances useful for business insights.

   Architecture configuration: We used 300 trees (n_estimators) during training, with a default depth parameter (a full-grown tree without pruning) and trained it on a SMOTE balanced data.

c) **XGboost (Model 2):** The final model was XGBoost, which is widely used in modern financial modeling because it handles, class imbalance (through the parameter: scale_pos_weight) and understands complex nonlinear relationships.

   Architecture configuration: XGBoost's architecture has regularization (L1 and L2 penalties) to control model complexity, shrinkage via learning rate, and column and row subsampling to reduce overfitting and improve generalization. In this model, learning was controlled using a learning_rate = 0.07 with a fixed ensemble size of 250 trees, allowing the model to incrementally refine predictions while reducing the risk of overfitting.

5. Training algorithm Pipeline:

Input: Dataset $\mathcal{D} = (X, y)$, where $y \in \{0,1\}$

1. Split $\mathcal{D}$into training and test sets. (X_{train}, y_{train}) and (X_{test}, y_{test}) using stratification to preserve class proportions.

2. Fit a scaler on X_{train} and apply it to both X_{train} and X_{test}

3. Handle class imbalance by applying SMOTE to (X_{train} , y_{train}) only for Baseline and Random Forest, additionally set a class weight w_1 for XGBoost, based on the imbalance ratio.

4. Train the model $\mathcal{M}$ on the balanced training data.

5. Evaluate $\mathcal{M}$ on $(X_{\text{test}}, y_{\text{test}})$ using ROC-AUC, precision, recall, F1-score, the confusion matrix, and probability-based ROC analysis.

6. Model Evaluation:

Logistic Regression demonstrated strong overall accuracy (94%) and solid discriminatory ability (ROC-AUC ≈ 0.926). However, performance on the **high-income** class was modest and only with a precision of **0.54** and recall of **0.46**. This means the model correctly identifies less than half of high-income individuals, though it rarely misclassifies low-income customers.

Random Forest offered the most balanced performance across metrics, achieving **95%** accuracy and a ROC-AUC of **0.936 (Fig. 5).** It improved precision for high-income customers **(0.63)** while maintaining similar recall to logistic regression (0.46). The model captures nonlinear relationships between demographic and labor-market variables, and its feature importance outputs remain relatively interpretable. This makes Random Forest especially useful for *segmentation and marketing use cases* where predictive strength and explainability coexist.

XGBoost achieved the highest ROC-AUC **(0.941)** and captured almost all high-income individuals, with a recall of **0.95**. However, its precision for the high-income class was low **(0.21),** meaning it generates many **false positives,** and it will flag numerous low-income customers as high-income. This behavior is expected from boosting models that are optimized for high recall. In my opinion, XGBoost is therefore ideal for top-of-funnel marketing, wealth-management outreach or early-stage lead expansion where "missing a high-value customer" is more costly than reviewing extra candidates.

Model usage recommendations:

- Start with **XGBoost** for ranking and prospect/customer scoring.
- Use **Random Forest** to identify key drivers for customer segmentation and marketing strategy.
- Use **Logistic Regression** where transparency and auditability are required.

Table 1 presents a comprehensive analysis of the results obtained through number of classification models. *More detailed classification report numbers are in the GitHub file.*

Table 1. Evaluation metrics for Classification models

Model	ROC-AUC	Precision (High-Income)	Recall (High-Income)	F1-Score (High-Income)	Overall Accuracy	Key Strength
Logistic Regression	0.926	0.54	0.46	0.5	0.94	Highly interpretable, regulatory-friendly
Random Forest	0.937	0.63	0.46	0.53	0.95	Strong accuracy + feature interpretability
XGBoost	0.941	0.21	0.95	0.34	0.77	Best recall and has excellent ranking power

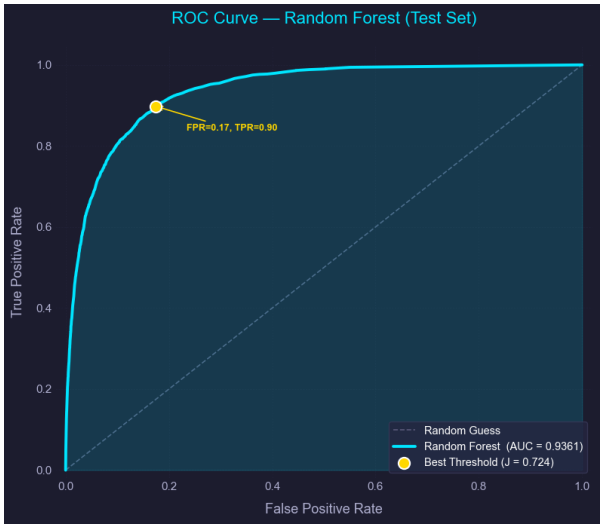

Figure 5. ROC curve for Random Forest on unseen Test set (AUC = 0.94)

Feature importances (**Fig. 6**) from Random Forest (best performing model) aligned strongly with our EDA. For example, **education level, occupation type, weeks worked in a year and age**. These results reinforce the consistency of our data-driven EDA. These variables exhibit strong correlation with earnings, and tree-based models are naturally effective at capturing such interactions.

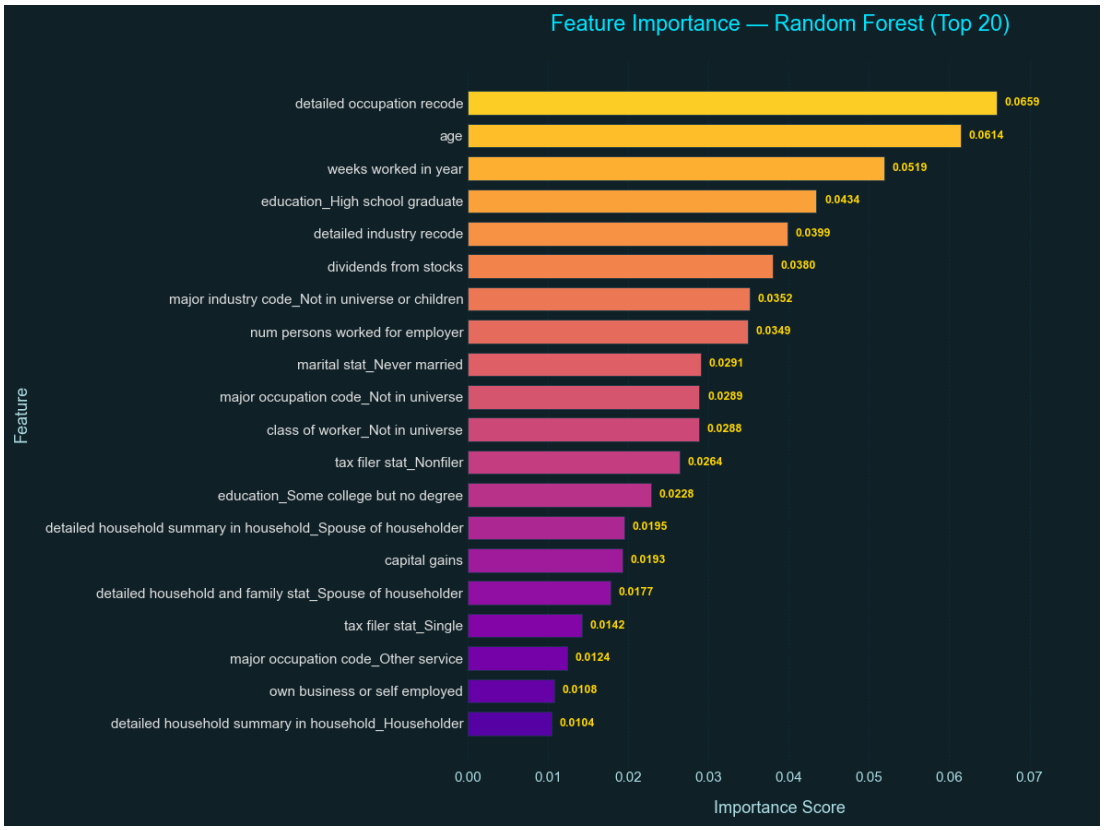

Figure 6. Feature importance score by Random Forest

7. Customer Segmentation Model:

To support the retail banking client's requirement a segmentation model was developed using the K-Means clustering algorithm. K-Means was chosen because of its strong performance on large, structured tabular datasets and its interpretability. My objective was not only to cluster the dataset statistically, but to create **stable, economically meaningful customer groups** that the JPMC risk teams could operationalize.

Before running the clustering model, few preprocessing steps were applied to ensure that the resulting segments reflected behavioral difference. **First**, individuals classified as "Children" (age <18) were **removed.** This group consistently formed its own cluster with zero weeks worked and no income. Including them artificially distorted cluster boundaries and consumed one full cluster regardless of the choice of *k*. Their removal ensured that the segmentation focused on economically active adults**,** which aligns with real-world business applications. **Second**, rather than performing clustering on the full set of 80+ census variables, the model used a small number of high-signal features derived from earlier weighted EDA. Variables such as *age, weeks worked per year, education level, capital gains, and marital status* demonstrated strong associations with income. We avoided the curse of dimensionality**,** which occurs when distance measures lose significance in very high-dimensional space.

Optimal number of cluster (K=4): The Silhouette Score was chosen because it measures both cluster cohesion and separation and showed that k = 5 in **Fig. 7** (left) achieved the highest numerical value (≈0.41). The Elbow Method in **Fig. 7** (right) provided additional support, as inertia declined steeply up to k = 4 and then flattened, suggesting that further increases in k yielded only marginal improvement in cluster compactness.

However, after examining the cluster assignments, the five-cluster solution produced two very similar midcareer, moderate-income groups that differed only slightly in work intensity. These groups offered no meaningful differentiation for marketing or product strategy. In contrast, k = 4 produced cleaner, economically distinct, and more actionable segments, separating (1) young low-income workers, (2) prime-age high-income professionals, (3) near-retirement individuals with limited work hours, and (4) stable mid-career workers with moderate earning potential. Although *k*=5 showed a marginally higher silhouette score, **k = 4 was selected as the final model because it delivered superior interpretability** which is the central goal of segmentation in a financial-services context.

The final clusters (**Table 2**) reveal clear differences in age, labor participation, education level, and income patterns. These differences translate directly into distinct customer personas with different needs, behaviors, and product affinities—forming the basis for targeted marketing, differentiated service models, and more accurate customer value assessments.

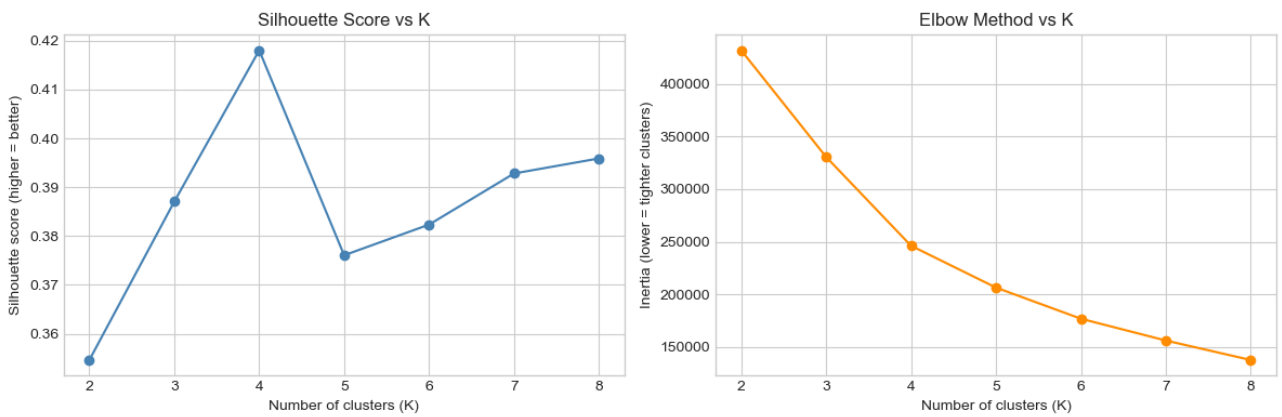

Figure 7. Silhouette Score for different k values

Table 2. Cluster analysis based on input features and probability of income >50K

cluster	age_mean	age_median	weeks_worked_mean	weeks_worked_median	education_mode	marital_status_mode	occupation_mode	income_binary_mean
0	26.225777	24.0	9.436359	0.0	High school graduate	Never married	Not in universe	0.003116
1	46.869231	46.5	47.458974	52.0	Bachelors degree(BA AB BS)	Married-civilian spouse present	Professional specialty	0.882051
2	68.766165	69.0	2.723369	0.0	High school graduate	Married-civilian spouse present	Not in universe	0.021713
3	39.638089	39.0	49.564557	52.0	High school graduate	Married-civilian spouse present	Professional specialty	0.132385

8. Business Recommendations:

1. Cluster 0: Retired / Non-Working Seniors (avg age 69, income >50K: 2.2%): This group is the oldest in the dataset with a median age of 69 and near-zero weeks worked (median 0). Their occupation is "Not in universe," confirming they are largely outside the labor force. Despite being married, their income probability is just 2.2%,

which is consistent with fixed-income retirement. They differ from Cluster 2 entirely by age, both groups are non-working, but Cluster 0 has exited the workforce through retirement while Cluster 2 has not yet entered it.

2. Cluster 1 - Mid-Career Professionals (avg age 40, income >50K: 14.3%): Full-year workers (median 52 weeks) in professional specialty occupations. They are typically married and in their late 30s to early 40s. They have the highest income probability of all four clusters at 14.3%, driven by their occupation type. What distinguishes them from Cluster 3 is not age or work hours, both groups are similarly aged and work similar weeks, but occupation. Professional specialty roles carry significantly higher earnings than administrative support roles.

3. Cluster 2 - Young Adults / Early-Career Individuals (avg age 26, income >50K: 0.3%): The youngest segment with a median age of 24, typically never married, with near-zero weeks worked (median 0) and "Not in universe" occupation status. This group likely includes students, early jobseekers who have not yet entered the workforce. Their income probability of 0.3% is the lowest across all clusters, reflecting the absence of earned income. They differ from Cluster 0 in that non-work here is a starting point, not an endpoint.

4. Cluster 3 - Working-Class Mid-Career Adults (avg age 39, income >50K: 6.3%): Like Cluster 1, these are full-year workers (median 52 weeks) in their late 30s, married, and economically active. The critical difference is occupation. Administrative and clerical roles carry a much lower earning premium than professional specialty. At 6.3% income probability, less than half of Cluster 1. This group represents the working middle class whose labor market participation is strong but whose occupational ceiling limits upward income mobility.

9. Future Work:

As a future extension of this work, I believe incorporating causal inference would help uncover *why* specific demographic patterns drive income outcomes. Additionally, evaluating fairness and bias is essential as income-based models risk reinforcing existing socioeconomic disparities. Integrating fairness metrics and mitigation strategies would be a necessary step before any real-world deployment to ensure the model supports equitable and responsible financial decision-making.

10. Conclusion:

In conclusion this analysis demonstrates that structured demographic and employment data can be operationalized to support income-based risk decisioning at JPMorgan Chase. From a risk perspective, the core finding is that income tier is highly predictable. XGBoost achieved an AUC of 94.1%, with Logistic Regression and Random Forest at 92.6% and 93.6% respectively. Education level, occupation type, and weeks worked in year emerged as the dominant risk signals, consistent across both weighted EDA and model feature importance rankings. This means the risk team can rely on a small, interpretable set of observable attributes to estimate a customer's income capacity without requiring self-reported income, which is often unreliable or unavailable at the point of application.

For the risk team, the immediate applications are threefold, **first**, the predictive model can serve as an income proxy layer in credit underwriting pipelines, flagging applicants where income capacity is uncertain. **Second**, the segmentation framework provides a basis for risk-adjusted product eligibility thresholds, ensuring that lending exposure is calibrated to each segment's income stability. **Third**, the segment-level income probabilities can inform stress testing assumptions, where Clusters 0 and 2 should be modelled with higher income shock sensitivity than Cluster 1. This work is a deployable analytical foundation and should be extended by incorporating bureau data, delinquency history and product utilization signals to move toward a complete risk-adjusted customer view.

12. References:

[1] https://www.census.gov/programs-surveys/cps.html
[2] https://scikit-learn.org/stable/
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[5] Grus, Joel. *Data science from scratch: first principles with python*. O'Reilly Media, 2019.